How-To Library

Informatica

Implementing Informatica Big Data Management on Oracle Big Data Cloud Service

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Abstract

You can deploy the Informatica Big Data Management solution on Oracle Big Data Cloud Service. This article describes the steps that you can use to implement Big Data Management on Oracle Big Data Cloud Service that uses a Cloudera CDH cluster with Kerberos, KMS, and SSL authentication enabled.

Supported Versions

- Informatica Big Data Management 10.1.1 HF1
- Informatica Big Data Management 10.2

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Overview

Customers of Informatica can deploy Big Data Management on Oracle Big Data Cloud Service.

Scenario

Oracle Big Data Cloud Service runs on Oracle Linux and the setup uses Cloudera CDH cluster with Kerberos, KMS, and SSL authentication enabled.

Pre-Implementation Tasks

Before you install Big Data Management, install and configure the Informatica domain and clients.

Perform the following tasks when you want to deploy Informatica Big Data Management 10.1.1 HF1 and 10.2:

1. To make the Informatica installer compatible with Oracle Linux, change properties in the `supported_os_kernel.ver` file in the `/Messages` location before you run install.sh
   Change the following lines:

   ```
   REDHAT_MIN_VERSION=32
   REDHAT_PATCH_VERSION_MIN_18=431
   ```

   to

   ```
   REDHAT_MIN_VERSION=39
   REDHAT_PATCH_VERSION_MIN_18=39
   ```
2. Run install.sh and complete the installation.

Implementation Tasks

Perform the following tasks when you deploy Informatica Big Data Management 10.1.1 HF1:

1. Install Big Data Management as specified in the pre-implementation tasks.

2. Configure Big Data Management.
   Before you run BDMConfig.sh from <INFA_HOME>/tools/BDMUtil, perform the following steps:
   1. Download the `connection.sh` file from the following location:
      `https://kb.informatica.com/h2l/HowTo%20Library/1/1105-BDMonOracleBigDataCloudService-Connection.zip`
   2. Extract the connection.sh file from the .zip file.
   3. Place the connection.sh file in the following location:
      `<INFA_HOME>/tools/BDMUtil/properties`
   4. Download the Hive client configuration from the CDH cluster and place it in following location:
      `<INFA_HOME>/tools/BDMUtil/properties/hive-conf`
   5. Run BDMConfig.sh with SSH option instead of the Cloudera Manager option.

Post-Implementation Tasks

Perform the following tasks when you deploy Informatica Big Data Management based on the version you install.

Informatica Big Data Management 10.1.1 HF1

1. Configure Big Data Management to run mappings in Hadoop environments.
   Verify that `<hostname>` in the `Metadata Connection String` property of the Hive connection is updated with the MetaStore server host name of the CDH cluster.

   Example:

   ```
   jdbc:hive2://<MetaStore server host name in CDH cluster>:10000/default;ssl=true;
   principal=hive/_HOST@BDACLOUDSERVICE.ORACLE.COM
   ```
2. If the CDH cluster is configured for high availability, copy the `core-site.xml` file from the cluster name node to the following location:
   `INFA_HOME/services/shared/Hadoop/<Hadoop distribution directory>/conf`
3. For mappings than run on the Spark engine, locate the `core-site.xml` in the following location:
   `INFA_HOME/services/shared/Hadoop/<Hadoop distribution directory>/conf`

   Comment out the following property:

   ```
   <property>
       <name>hadoop.security.key.provider.path</name>
       <value>kms://https@cfclbv1891.us2.oraclecloud.com;cfclbv1892.us2.oraclecloud.com:
   16000/kms</value>
     </property>
   ```

Informatica Big Data Management 10.2

1. For mappings that run on the Spark engine, add the hadoop.rpc.protection property to the `hive-site.xml` file in the in cluster configuration object and specify the value as `privacy`
2. If the CDH cluster has Sentry authorization enabled, assign required permissions to the Spark staging directory that is specified in the Hadoop connection.

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